Example (Bernoulli) $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$

$$\mathcal{L}(p; x) = \prod_{j=1}^{n} p^{x_j}(1-p)^{1-x_j}, \quad p \in [0,1]$$

$$\ell(p) = \sum_{j=1}^{n} \{x_j \log(p) + (1-x_j)\log(1-p)\}$$

$$= \log(p)\left(\sum_{j=1}^{n} x_j\right) + \log(1-p)\left(\sum_{j=1}^{n}(1-x_j)\right)$$

$$\ell'(p) = \frac{1}{p} S - \frac{1}{1-p}\cdot(n-S), \quad S = \sum_{j=1}^{n} x_j$$

Setting $\ell'(p)$ equal to 0,

$$\frac{S}{p} = \frac{(n-S)}{1-p}$$

$$\Rightarrow S(1-p) = (n-S)p$$

$$\Rightarrow S = np$$

$$\Rightarrow \hat{p} = S/n = \bar{x}$$

Example (Weibull)

$$f(x; \lambda, k) = \left(\frac{k}{\lambda}\right)\cdot\left(\frac{x}{\lambda}\right)^{k-1} \exp\left(-\left(\frac{x}{\lambda}\right)^{k}\right)$$

$$x > 0, \quad k, \lambda > 0$$

Suppose $X_1, \ldots, X_n$ is a random sample from this distribution

$$\mathcal{L}((\lambda, k)) = \prod_{j=1}^{n} \left(\frac{k}{\lambda}\right)\cdot\left(\frac{x_j}{\lambda}\right)^{k-1} \exp\left(-\left(\frac{x_j}{\lambda}\right)^{k}\right)$$

log likelihood

$$\ell(\lambda, k) = \sum_{j=1}^{n}\left\{\log\left(\frac{k}{\lambda}\right) + (k-1)\log\left(\frac{x_j}{\lambda}\right) - \left(\frac{x_j}{\lambda}\right)^{k}\right\}$$

$$= n\log\left(\frac{k}{\lambda}\right) + (k-1)\sum_{j=1}^{n}\log\left(\frac{x_j}{\lambda}\right) - \sum_{j=1}^{n}\left(\frac{x_j}{\lambda}\right)^{k}.$$

Let $f(\underline{x};\theta)$ be the joint pdf of $X = (X_1, \ldots, X_n)$ so the log likelihood is

$$\ell(\theta; \underline{x}) = \log f(\underline{x};\theta) \qquad \theta \in \Theta$$

(Fisher) Score function is

$$U(\theta) = U(\theta; \underline{x}) = \frac{\partial}{\partial\theta} \ell(\theta; \underline{x}) \qquad \theta \in \Theta$$

Assuming we can interchange differentiation wrt θ and integration wrt x twice, so

$$\int \frac{\partial}{\partial\theta} f(\underline{x};\theta)\, d\underline{x} = \frac{\partial}{\partial\theta} \int f(\underline{x};\theta)\, d\underline{x}$$

Then $$\mathbb{E}_\theta[U(\theta;X)] = \mathbb{E}_\theta\left[\frac{\partial}{\partial\theta} \log f(\underline{X};\theta)\right]$$

CB eq 7.3.7-8 $$= \int \frac{1}{f(\underline{x};\theta)} \left\{\frac{\partial}{\partial\theta} f(\underline{x};\theta)\right\} f(\underline{x};\theta)\, dx$$

$$= \int \frac{\partial}{\partial\theta} f(\underline{x};\theta)\, dx = \frac{\partial}{\partial\theta} \int f(\underline{x};\theta)\, dx = \frac{\partial}{\partial\theta}[1] = 0$$

The quantity $\operatorname{var}_\theta[U(\theta;X)] = \mathbb{E}_\theta[U^2(\theta;X)] =: I(\theta)$ is called the expected (Fisher) information.